Lab 2 IoT Security

ESE5180: IoT Wireless, Security, & Scaling

Group Number:

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GitHub Repository URL: https://github.com/Rico-Zhuang/ESE5180_Lab2_Fall2026_Rico_Talal

Part 1 — Implementing Our Own LoRa Security

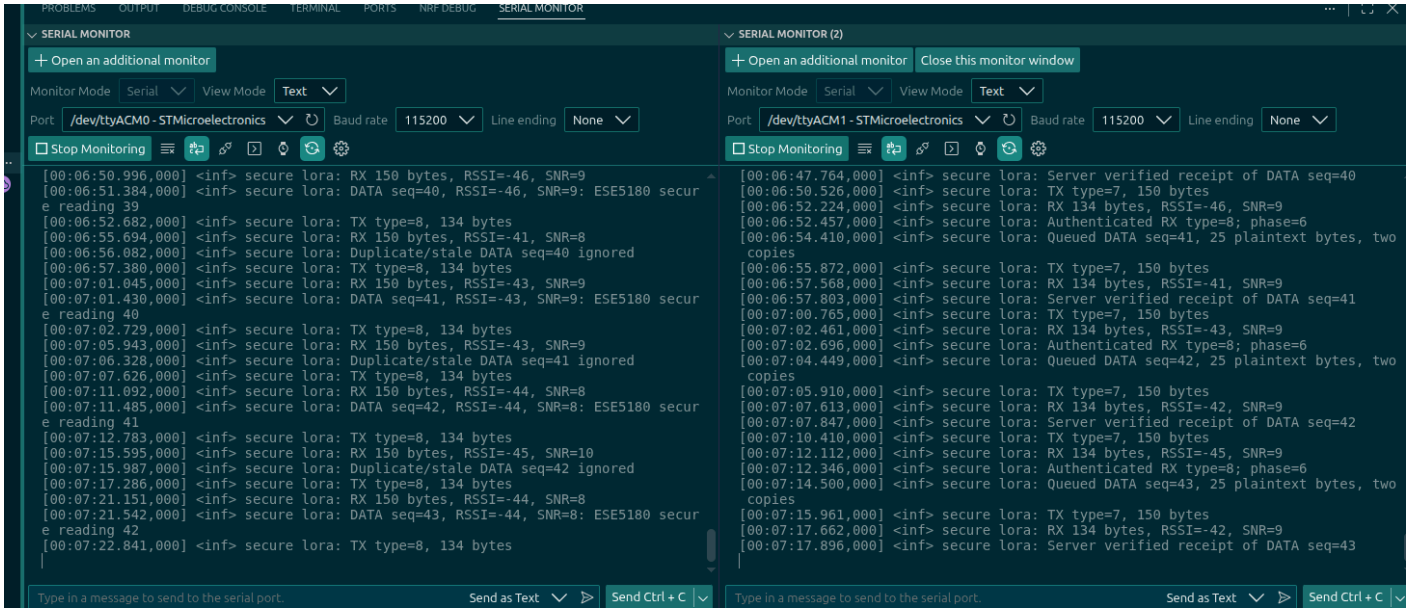
The client and server share the protocol code. See build and checkoff instructions.

Both firmwares passed hardware checks for pairing, encrypted delivery, duplicate rejection, receipts, and timeout recovery; all ten host test groups passed. The teaching-team checkoff is still required. Builds and private signing keys are excluded from Git.

Answer 1.1

We chose Curve25519 Diffie–Hellman because the starter supports it and it creates fresh shared keys without transmitting a secret. Ed25519 signatures authenticate peers, and HKDF-SHA256 derives the AES key; this requires provisioning identities and a larger handshake.

AES provides confidentiality, while signatures and replay checks protect integrity. Retries and timeout recovery support availability during packet loss. Jamming, traffic analysis, and physical key extraction remain risks, and LoRa's range lets attackers operate farther away. Occasional lost readings may be acceptable for a noncritical sensor, but availability against jamming is not guaranteed.



Part 2 — Encryption Speed and Memory Tradeoffs

All four benchmark variants ran on the same NUCLEO-WL55JC at 48 MHz, using 200 iterations and a 176-byte message. Each passed the known-answer and round-trip checks. Saved measurements were collected on October 26, 2026.

Answer 2.1 — Speed

Variant	Key setup, first call (cyc)	Key setup (cyc)	Encrypt (cyc/B)	Decrypt (cyc/B)	Encrypt 176 B (µs)
SW, RAM tables	46,319	8,815	82	82	301.628
SW, ROM tables	8,891	8,886	106	106	392.251
SW, ROM + fewer	8,902	8,882	107	107	395.003
HW	1,098	1,036	62	81	228.128

Cycle averages are rounded to integers; microseconds use the full accumulated cycle count.

Answer 2.2 — Memory

Variant	text	data	bss	Flash (text + data)	RAM (data + bss)
SW, RAM tables	30,704	220	19,028	30,924	19,248
SW, ROM tables	38,912	220	10,280	39,132	10,500
SW, ROM + fewer	32,808	220	10,280	33,028	10,500
HW	26,464	492	9,304	26,956	9,796

Values are bytes from `arm-zephyr-eabi-size`.

Answer 2.3 — Locating the tables

Variant	FT0 address	FT0 in RAM or flash	FT1 present	FSb address
SW, RAM tables	`0x20002510`	RAM	yes	`0x20002c24`
SW, ROM tables	`0x080089f0`	flash	yes	`0x080096fd`
SW, ROM + fewer	`0x08007210`	flash	missing	`0x08007f25`
HW	missing	missing	missing	missing

Answer 2.4 — Choosing a variant

We would choose hardware AES: it used 9,796 bytes of RAM versus 19,248 for RAM tables. It also encrypted 176 bytes in 228.128 µs versus 301.628 µs. With encryption only once every ten minutes, saving scarce RAM matters more than the timing difference.

Part 3 — Hardware Unique Keys

The implementation and reproduction steps use the starter sample key, as requested, instead of the unavailable Lab 1 key.

Answer 3.1 — Plaintext storage

The 16-byte sample key is visible at `0x0803F000` in Board A's NVS dump.

```
rico@rico-LOQ-15ARP10E:~/ESE5180/ESE5180_Lab2_Fall2026_Rico_Talal/lab2_security/Hardware_Unique_Keys/bui
ld_evidence$ xxd -g 1 -c 16 -l 128 -o 0x0803F000 nvs-plaintext.bin
0803f000: 2b 7e 15 16 28 ae d2 a6 ab f7 15 88 09 cf 4f 3c  +~..(.........O<
0803f010: ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff  ................
0803f020: ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff  ................
0803f030: ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff  ................
0803f040: ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff  ................
0803f050: ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff  ................
0803f060: ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff  ................
0803f070: ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff  ................
```

Answer 3.2 — Encrypted storage

HKDF-SHA256 derives an AES-128 key from the board UID and application salt. NVS stores the IV, CBC ciphertext, and a digest for checking decryption. Serial provisioning keeps the plaintext key out of the firmware; it was absent from the entire 256 KiB flash dump.

```
ld_evidence$
rico@rico-LOQ-15ARP10E:~/ESE5180/ESE5180_Lab2_Fall2026_Rico_Talal/lab2_security/Hardware_Unique_Keys/bui
ld_evidence$ xxd -g 1 -c 16 -l 128 -o 0x0803F000 nvs-encrypted.bin
0803f000: 49 4f 8e a8 67 1b 0f 78 d9 cc a6 08 ad e7 64 ac  IO..g..x......d.
0803f010: 4e 8f d7 cf eb e9 39 63 19 9d 22 f6 74 82 76 7a  N.....9c..".t.vz
0803f020: d4 ff b8 b7 7f 7d 6b 26 19 6e 9a 07 0e 98 3f 67  .....}k&.n....?g
0803f030: 01 a4 c4 2d ec 81 3d 4d e1 a5 35 d2 0a 7d f5 36  ...-..=M..5..}.6
0803f040: ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff  ................
0803f050: ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff  ................
0803f060: ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff  ................
0803f070: ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff  ................
```

Answer 3.3 — Cloning

Board A decrypted successfully. Board B rejected the identical 256 KiB flash image because its UID differs; both results persisted after reset. Serial output and flash comparison record the test. Both boards remain ready for the teaching-team demonstration.

This demonstrates rejection of an exact clone. The UID is readable, so it is not a secret HUK; the CBC format also does not authenticate arbitrary record changes.